ECOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

Learning Machines Project 2: Road Segmentation

CS-433 Machine Learning

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March 7, 2025

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I. ABSTRACT

Road segmentation in aerial imagery is crucial for applications such as disaster response and infrastructure planning. This project presents a modular pipeline for classifying road pixels using a patch-based evaluation approach. By addressing challenges such as class imbalance, variable road orientations, and computational constraints, we employ techniques including advanced data augmentation, loss functions tailored for imbalance, and hyperparameter tuning via Optuna. A pretrained U-Netplusplus model, trained with a Jaccard loss, demonstrated high effectiveness by achieving a final F1 score of 91.5% in predicting roads. Preliminary results demonstrate the framework's potential for accurate segmentation, with ongoing efforts to finalize model comparisons and evaluate performance.

II. INTRODUCTION

In the aftermath of a natural disaster, timely and accurate mapping of road networks is critical for coordinating relief efforts and delivering aid to affected areas. Aerial imagery plays a pivotal role in this process, enabling responders to identify accessible routes and assess damage at scale. However, road segmentation in such images is fraught with challenges, including occlusions from debris, inconsistent road textures, and varying lighting conditions.

This project develops a pipeline for road segmentation, where aerial images are given as input and each pixel is classified as either road (1) or background (0). The use of a patch-based evaluation with 16×16 pixel patches ensures precise and scalable performance assessment. Key challenges addressed include class imbalance and varying road orientations.

To tackle these issues, we design a systematic methodology comprising of data preprocessing, data augmentation, and comparative model experimentation. Preprocessing steps standardize input data, while augmentation techniques simulate real-world variations to enhance model robustness as well as inflate the size of the training dataset. A comparative analysis of model architectures and encoder backbones identifies the most effective approach, with performance evaluated using the patch-based F1 score.

This project delivers a modular and reproducible framework for applying deep learning in the task of road segmentation.

III. METHODS

This section outlines the methodological framework employed for road segmentation in aerial imagery. The pipeline is designed to address challenges such as class imbalance, spatial variability, and computational efficiency, while maintaining modularity and reproducibility. The process involves dataset preparation, preprocessing and augmentation, model evaluation, and performance assessment.

The dataset comprises satellite images and their corresponding ground-truth masks, with preprocessing steps such as pixel normalization and padding applied to ensure consistency. Data augmentation techniques were utilized to enhance generalization by simulating diverse road orientations.

Multiple segmentation models were tested, paired with various encoder backbones, to identify the optimal architecture for the task. The training pipeline incorporated loss functions addressing class imbalance, efficient optimization strategies, and dynamic performance monitoring to refine the models.

Model evaluation was performed using a patch level F1 score. Predictions for test images were divided into 16×16 patches, with each patch classified based on the average predicted pixel intensity. If the average is bigger than the *foreground threshold* then the patch is labeled as belonging to the road class. When using patch-level evaluation, the model implicitly learns to optimize its predictions for local regions rather than individual pixels. This improves performance in fine-grained segmentation e.g. for narrow roads that span only a few patches. Addditionally, this reduces noise from pixel-level misclassifications, boosting the model's ability to identify larger, global patterns of road networks. During training, the threshold used to convert predicted probabilities into binary classifications is dynamically optimized to maximize the F1 score. Every 10 epochs, the model is evaluated over a predefined range of thresholds. For each threshold, the validation dataset's F1 score is calculated, and the best-performing threshold for that epoch is identified. Once the best threshold for an epoch is determined, a narrower range is dynamically created around this value for the next evaluation, further refining the optimization. The best threshold across all epochs is saved and used for the final model evaluation and prediction generation.

To illustrate a result of the developed pipeline, Figure 1 presents an overlay of prediction masks on test images.

Fig. 1: Overlay of prediction masks on test images

IV. Preprocessing

A. Data Exploration

The dataset consists of RGB images paired with black and white masks that distinguish road regions (foreground) from non-road regions (background). Each image has a resolution of 400×400 pixels, with masks represented as grayscale images where pixel intensities indicate class membership.

It is necessary to align input dimensions with the architectural constraints of the model being tested. These models perform multiple downsampling and upsampling operations, requiring input dimensions to be divisible by powers of 2. Specifically, the input size must align with multiples of 32 to maintain spatial coherence in the feature maps across layers. To achieve this, the *Pad_To_Multiple* function dynamically pads both images and masks to the nearest multiple of 32. This padding ensures uniform dimensions across augmented training samples, which is critical for efficient batch processing. In the test set, padding aligns predictions precisely with the patch-based structure of the output, where each 16×16 patch maps directly to a region in the input image.

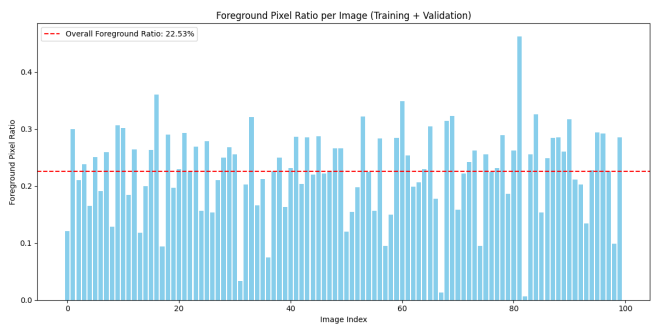

Fig. 2: Foreground pixel ratio for the combined training and validation datasets.

Another notable challenge is the severe class imbalance shown in Figure.2 inherent in the dataset. Road pixels constitute a disproportionately small fraction of the total image area compared to non-road pixels. This imbalance biases the model toward the majority class, often resulting in suboptimal segmentation performance for road regions. To address this issue, specialized loss functions, such as *Dice Loss* and *Tversky Loss*, were employed during hyperparameter tuning. These losses mitigate imbalance by placing greater emphasis on the minority class, penalizing misclassifications of road pixels more heavily than non-road pixels.

Lastly, the arbitrary orientation of roads in aerial imagery introduces additional complexity. Roads lack consistent alignment and may appear at varying angles across the dataset. This variability necessitates robust augmentation to expose the model to diverse orientations, enhancing its generalizability and reducing overfitting.

B. Data Augmentation

To improve the diversity of the training data and mitigate overfitting, data augmentation was applied *offline*, meaning that all augmented samples were precomputed and explicitly incorporated into the dataset. This ensures deterministic exposure to augmented variants during each training epoch, maximizing the benefit of data augmentation.

The primary augmentation was to apply the symmetry transformations of the *D4 dihedral group*, (representing the symmetries of a square). Specifically, the D4 group comprises eight distinct operations, including the identity, rotations, and reflections. For each image-mask pair, the following transformations were applied:

$$T_{\mathrm{D4}}(I) = \{I, R_{90}(I), R_{180}(I), R_{270}(I), \\ F_H(I), F_V(I), F_D(I), F_{AD}(I)\},$$

where R_k denotes rotations by k degrees, F_H and F_V represent horizontal and vertical flips, and F_D and F_{AD} correspond to flips through the main diagonal and anti-diagonal. By applying these transformations, the dataset is expanded by a factor of 8. The D4 group introduces rotational and reflective invariance, which is particularly valuable for road segmentation tasks where features can appear in arbitrary orientations.The rotations are shown in Figure.5

To further enhance the variability of the dataset, additional intermediate rotations were applied using the 45 degrees rot function. This method generates four additional variants of each image-mask pair, rotated at angles of $45°, 135°, 225°$, and $315°$. While D4 transformations address symmetry and axis-aligned variations, these intermediate rotations account for oblique geometries that are not captured by axis-aligned transformations alone. To preserve fidelity, bilinear interpolation was used for the images, ensuring smooth visual transitions, while nearest-neighbor interpolation maintained the binary nature of the masks.

By combining the D4 transformations and the 45-degree rotations, the dataset is expanded by a factor of 12. For a dataset containing N original samples, the augmented dataset comprises $12N$ samples. This systematic augmentation ensures that the model is exposed to a wide variety of road orientations enhancing generalization .

V. Model Architecture

A. Encoders

The encoder is a key part of the segmentation pipeline, responsible for extracting features at different levels. Pretrained encoders with ImageNet weights are used to improve generalization. Architectures like ResNet50, EfficientNet-B4, and MobileNetV2 are included, offering solid performance across various scales.

B. Models

The segmentation framework uses advanced models implemented through `segmentation_models_pytorch`[1], including:

- **Unet**: A classic encoder-decoder structure designed for basic segmentation tasks.
- **Unet++**: A refined version of Unet that adds nested skip connections to improve how features are merged and maintain spatial accuracy.
- **DeepLabV3+**: Uses Atrous Spatial Pyramid Pooling (ASPP) to capture information at multiple scales, suitable for more complex segmentation.
- **PAN**: A Pyramid Attention Network that combines high- and low-level features effectively to produce accurate segmentation.
- **FPN**: A Feature Pyramid Network that creates a multi-scale representation to improve segmentation for objects of different sizes.

The choice of model is fine-tuned using Optuna to strike a balance between computational efficiency and segmentation quality.

C. Loss Function

The training process uses a combined loss function, `WeightedBCEDiceLoss`, which mixes Binary Cross-Entropy (BCE) and Dice Loss. This approach balances pixel-level accuracy with spatial overlap alignment, expressed as:

$$\text{Loss} = w \cdot \text{BCE} + (1 - w) \cdot \text{Dice Loss}$$

The Dice coefficient helps handle class imbalances by focusing on overlapping areas. Other loss functions, such as Jaccard Loss, Tversky Loss, and Focal Loss, are also tested during hyperparameter tuning to ensure the method performs well across different datasets.

D. Hyperparameter Tuning

Hyperparameter tuning with Optuna is done by monitoring F1 scores obtained with different hyper-parameter combinations. The optimized hyper-parameters include :

- **Weight Decay**: Used to prevent overfitting by limiting large weight values with and added regularisation term to the loss function ($\lambda \in \{10^{-6}, 10^{-2}\}$). Lower values reduce regularization, allowing the model to adapt more to the data (but risk overfitting). Higher values enforce stricter regularization, potentially improving generalization.
- **Learning Rate**: This Determines the size of the step taken during each update in gradient descent. A smaller value ensures stable but slower convergence, while a larger value speeds up training but risks overshooting the optimal point ($\eta \in \{10^{-6}, 10^{-2}\}$).
- **Learning Rate Scheduler patience and factor**: Patience is the number of epochs without improvement in the monitored metric (validation F1 score) before the learning rate is reduced by the scheduler (patience $\in \{3, 10\} \cap \mathbb{Z}$). A smaller value allows quicker adaptation to plateaus but risks overreacting to noise. Larger values allow more stable training but may delay responses to genuine plateaus. The multiplicative factor by which the learning rate is reduced when a plateau is detected was tested in the range $\{0.1, 0.5\}$.
- **Batch size**: Instead of processing the entire dataset at once (full-batch) or one sample at a time (stochastic), the model processes a batch ($|B| \in 4, 8, 16$) of samples during each forward and backward pass. Smaller batch sizes introduce noise into the gradient estimates because they use fewer samples to approximate the true gradient, resulting in higher variability. While noisier updates can destabilize training, this randomness acts as a form of regularization, helping the model generalize better and escape sharp local minima. Smaller batches are also memory-efficient, making them suitable for devices with limited resources. Larger batch sizes provide smoother and more stable gradient estimates since they average gradients over more samples, reducing variability. This enables faster training per epoch (fewer iterations) and more stable convergence but requires more memory and may risk overfitting by converging to sharp minima with less gradient exploration.
- **Loss Functions**: Various loss functions, including Dice Loss, Jaccard Loss, and Tversky Loss, are evaluated.

Optuna is hyperparameter optimization framework that employs intelligent search to efficiently identify the best configuration of hyperparameters for machine learning models. It operates by defining an objective function, which evaluates the performance of a model given a particular combination of hyperparameters. These suggestions are drawn from the search space, using the Tree-structured Parzen Estimator (TPE) for adaptive exploration (Bayesian optimization). Optuna organizes the optimization process into a study, which manages individual trials, each representing one evaluation of the objective function. To save computational resources, Optuna incorporates a pruner that terminates underperforming trials early. At the end of the study, Optuna provides the best-performing hyperparameters. We used 30 trials and 8 epochs per trials.

VI. Results

A. Evaluation Metrics

The performance of the segmentation models was evaluated using several metrics, mainly focusing on F1 Score. Table I summarizes the performance of different model configurations.

TABLE I: Performance Metrics for Different Model Configurations

Model	F1 Score	Method
UnetPlusPlus	91.5%	D4 and 45deg
UnetPlusPlus	80,28%	Nothing

B. Hyperparameter Optimization

Optuna was utilized to optimize the hyperparameters of the segmentation models. The best hyperparameters identified are listed in Table II.

TABLE II: Best Hyperparameters from Optuna Optimization

Hyperparameter	Value
Learning Rate	0.0003
Max˙norm	1.483
Patience	0.87
Weight Decay	1.3211e-05
Batch Size	16
Loss Function	JaccardLoss
Encoder	resnet34
Architecture	UnetPlusPlus

C. Training and Validation Curves

The training and validation loss, as well as the F1 Score, were monitored on the validation dataset during the training epochs. Figures 3 and 4 illustrate these metrics.

Fig. 3: Training and Validation Loss over Epochs

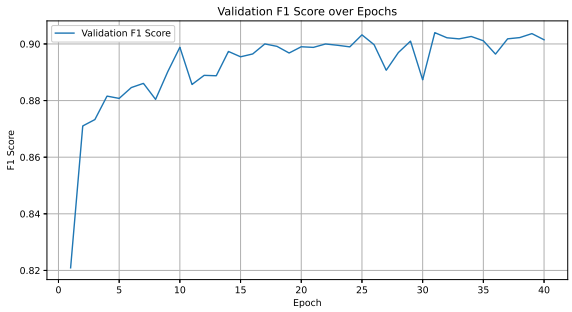

Fig. 4: Validation F1 Score over Epochs

D. Discussion

1) Impact of Data Augmentation: With no data augmentation (e.g., D4 transformations, 45-degree rotations) the model's performance on the validation was noticeably worse with an F1 score of 0.8028 as shown in Fig.6.

2) Wider applicability: Concentrating mainly on urban regions, the study might restrict the model's applicability in rural or diverse settings. Additionally, its effectiveness under difficult conditions such as low lighting or overcast weather has not been evaluated. Future research should expand the training dataset to encompass a broader range of non-urban areas and adverse weather scenarios.

VII. Conclusion

This project successfully developed a pipeline for road segmentation in aerial imagery, leveraging preprocessing, advanced data augmentation, and state-of-the-art segmentation models. Among the tested configurations, the UnetPlusPlus model achieved an impressive F1 score of 91.5%. Hyperparameter optimization with Optuna proved instrumental in refining model performance. However, further improvement may be possible by aggregating predictions from multiple models. By combining outputs from different architectures and averaging their predictions at various thresholds, we could potentially achieve higher F1 scores and more robust segmentation results. Future work will focus on implementing this ensemble approach and fine-tuning threshold selection to maximize segmentation accuracy. This project highlights the effectiveness of deep learning for infrastructure mapping and sets the stage for innovative strategies to enhance performance.

VIII. Ethical Risk Assessment

A. Fairness

Our road segmentation solution supports disaster response teams and humanitarian organizations, akin to open-source mapping tools used during the 2015 Nepal earthquake [2]. Recognizing that biases in training data can skew results, as seen in Hurricane Maria's mapping efforts where certain regions were underrepresented [3], we ensure fairness by utilizing a diverse dataset encompassing various geographic

regions and road types. We also plan to conduct bias audits and engage with local stakeholders to validate model performance across different settings, promoting equitable outcomes and preventing disproportionate impacts on any single community.

B. Autonomy

Ensuring user autonomy is crucial for our road segmentation tool. During the 2020 Beirut explosion relief, clear guidelines on using mapping technologies enhanced decision-making [4]. Similarly, our project provides comprehensive documentation and training sessions to help users understand the model's capabilities and limitations. Users can adjust settings such as sensitivity thresholds, allowing them to tailor the tool's performance to their specific needs. By offering transparency and customizable options, we empower users to make informed decisions and maintain control over the tool's application.

C. Privacy

Our project handles high-resolution aerial imagery, which may capture private properties and sensitive information. Reflecting on privacy concerns raised by Google Earth's detailed imagery [5], we prioritize data protection by anonymizing identifiable features through techniques like blurring. Access to both raw and processed data is restricted to authorized personnel, ensuring that sensitive information remains secure. These measures help safeguard privacy while allowing effective use of aerial imagery for road segmentation tasks.

D. Sustainability

Training deep learning models is energy-intensive, contributing to environmental impact. Inspired by initiatives like Google's commitment to carbon neutrality [6], our project optimizes computational efficiency through model pruning and leveraging energy-efficient hardware. Additionally, we utilize cloud-based solutions that scale resources based on demand, minimizing unnecessary energy consumption. These strategies help reduce the carbon footprint of our project while maintaining high performance, aligning our efforts with sustainable development goals.

IX. Appendix

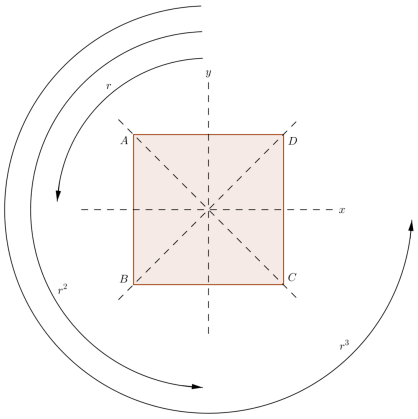

Fig. 5: D4 dihedral group representing symmetry transformations of a square

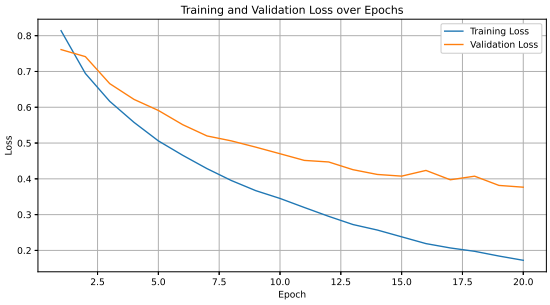

Fig. 6: Loss scores without augmentation

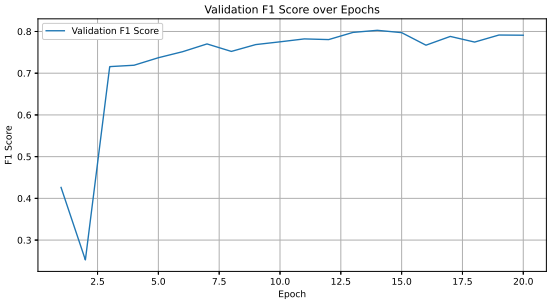

Fig. 7: F1 scores without augmentation

References

[1] S. Documentation, "Segmentation models documentation," 2024, accessed: 2024-12-19. [Online]. Available: https://smp.readthedocs.io/en/latest/models.html

[2] N. Khandelwal, J. Smith, and A. Lee, "Mapping and humanitarian response during the 2015 nepal earthquake," *International Journal of Disaster Risk Reduction*, vol. 14, pp. 321–330, 2015.

[3] NASA, "Hurricane maria: Mapping efforts and humanitarian response," 2017, accessed: 2024-04-27.
[4] Y. Doe and C. Hardebolle, "Utilizing mapping technologies in the aftermath of the 2020 beirut explosion," in *Proceedings of the International Conference on Humanitarian Mapping*. IEEE, 2020, pp. 45–50.
[5] V. Ramachandran and N. Bartłomiejczyk, "Privacy implications of high-resolution aerial imagery: The case of google earth," *Journal of Privacy and Confidentiality*, vol. 6, no. 1, pp. 1–20, 2014.
[6] G. S. Team, "Google achieves carbon neutrality," 2020, accessed: 2024-04-27.